

Tech Challenge FASE 5 — Hackathon SolidaryTech

Relatório de entrega final. Atende a todos os requisitos do PDF "Desafio - Tech Challenge Fase 5.pdf" (Hackathon de 2 meses, vale 90%).

Identificação do grupo

Participante	RM
Luiz Cesar Paulino da Costa	RM367925
Marcelo Oliveira	RM368372
Ricardo Pimenta	RM368911
Sergio Muller	RM368926

Turma: Pós-Graduação — DevOps & Arquitetura de Software — FIAP Pos Tech

Repositório: <https://github.com/marcelosilva-dev/tech-challenge-5-fiap> **Vídeos de demonstração:** pasta compartilhada no Google Drive — ver [seção dedicada](#) abaixo **Data de entrega:** 2026-07-17

Vídeos de demonstração

Gravação dividida em 5 partes. Pasta compartilhada (Google Drive):
https://drive.google.com/drive/folders/1Kp_Ar4HJBincMk72db34X-i-A9DyMWQT?usp=sharing

Ordem de visualização recomendada:

#	Arquivo	Conteúdo
1	01-overview-arquitetura.mkv	Pitch executivo + visão geral da arquitetura dos 3 microsserviços
2	02-github-actions-and-terraform.mkv	CI/CD com DevSecOps (GitHub Actions) + Infraestrutura como Código (Terraform)
3	03-gitops.mkv	GitOps em ação — ArgoCD sincronizando os Applications
4	04-sli-slo-sla-errorbudget.mkv	SRE — SLI/SLO/SLA, Error Budget e self-healing (redução de MTTR)
5	05-disaster-recovery.mkv	Disaster Recovery — Velero (backup) + Warm Standby cross-region

Sumário executivo

Este projeto constrói, orquestra, monitora, otimiza financeiramente e estabelece estratégia de resiliência para o ecossistema de microsserviços da **SolidaryTech** — plataforma sem fins lucrativos que conecta ONGs a doadores e voluntários.

3 microsserviços do upstream `dougl's/hackathon-DCLT` :

Serviço	Stack	Criticidade
ngo-service	Python/Flask + PostgreSQL	Importante
donation-service	Go + PostgreSQL + SQS	● Hot Path
volunteer-service	Python/Flask + DynamoDB	Importante

Todos rodam em **Amazon EKS** com: - ✓ IaC modular (Terraform 1.5+) - ✓ CI/CD com DevSecOps (GitHub Actions: Trivy + gosec + bandit + flake8 + golangci-lint) - ✓ GitOps (ArgoCD com auto-sync + self-heal) - ✓ Observabilidade total (Prometheus + Loki + Grafana + OpenTelemetry) - ✓ APM com Distributed Tracing (New Relic) - ✓ SRE com SLO/SLI/Error Budget - ✓ FinOps com tags rigorosas e forecast - ✓ ITSM com PagerDuty + Discord + Self-healing automatizado - ✓ Disaster Recovery cross-region (Warm Standby + DynamoDB Global Tables + Velero)

0. Fundação DevOps (Fases 1-4) — Requisito Obrigatório

Docker e Kubernetes

- **Dockerfiles otimizados multi-stage** em `microservices/<service>/Dockerfile` :
- **donation-service (Go):** `golang:1.26-alpine` (builder) + `gcr.io/distroless/static-debian12:nonroot` (runtime) — imagem final ~10MB, zero CVEs de SO
- **ngo + volunteer (Python):** `python:3.12-slim` com venv copiado entre stages, non-root user UID 10001, `PYTHONUNBUFFERED=1`
- **Deploy em EKS** (Amazon Elastic Kubernetes Service) — `terraform/modules/eks/`
- 3 ECR repositórios com `scan_on_push = true` e lifecycle policy (keep last 10)

Infraestrutura como Código

`terraform/` com 5 módulos reutilizáveis (`networking` , `eks` , `databases` , `messaging` , `ecr`) + 2 environments (`primary` em us-east-1, `dr` em us-west-2).

Backend remoto S3 + DynamoDB lock com nomes sufixados por `ACCOUNT_ID` para unicidade global.

CI/CD com DevSecOps

3 pipelines em `.github/workflows/ci-<service>.yaml` . Cada um:

Step	Ferramenta	Função
lint	flake8 / golangci-lint	Estilo de código
test	pytest / go test	Unit tests
SAST	bandit (Python) / gosec (Go)	Static Application Security Testing
SCA	Trivy filesystem scan	Software Composition Analysis
build	docker build multi-stage	Imagem amd64
container scan	Trivy	Scan da imagem final
push	docker push	ECR (tag = commit SHA + latest)
update GitOps	sed + git commit	Atualiza gitops/<svc>/deployment.yaml

GitOps

ArgoCD instalado em namespace `argocd`. 4 Applications declarativas em `argocd/applications.yaml` : - `solidarytech-shared` (namespace + Ingress) - `ngo-service`, `donation-service`, `volunteer-service`

Sync policy: `automated` com `prune: true` e `selfHeal: true`.

Observabilidade e APM

- **Métricas:** kube-prometheus-stack (Prometheus + Grafana + Alertmanager + node-exporter + kube-state-metrics)
- **Logs:** Loki (single-binary mode, 72h retention) + Promtail (DaemonSet)
- **Traces:** OpenTelemetry Collector como hub central, OTLP gRPC+HTTP
- **APM externo:** New Relic via OTLP HTTP exporter
- **Instrumentação:**
 - donation-service (Go): OTEL SDK manual com `otelhttp` middleware
 - ngo + volunteer (Python): `opentelemetry-distro` auto-instrumentation
- **Dashboard customizado:** Dashboards → SolidaryTech → SolidaryTech Overview

→ Detalhe técnico em [docs/SRE-SLO.md](#)

1. SRE — Confiabilidade e Golden Metrics

Definição de SLOs e SLIs

Documento completo em [docs/SRE-SLO.md](#). Resumo:

	donation-service (Hot Path)
SLI #1	Availability 5m (% requisições sem 5xx)
SLI #2	Latência P95 5m

	donation-service (Hot Path)
SLO Availability	≥ 99.9% (Error Budget: 43min/mês)
SLO Latência P95	< 300ms
SLA contratual ONGs	≥ 99.5% disponibilidade, P95 < 500ms

PrometheusRules em [gitops/monitoring/alerting/prometheus-rules.yaml](#) .

Dashboard SRE

Painel exclusivo em Grafana mostrando: - SLI Availability vs SLO 99.9% (com linha de threshold) - SLI Latency P95 vs SLO 300ms - Error Budget consumido nas últimas 24h / 7 dias / 30 dias - Logs do donation-service correlatos via Loki

MTTR — redução comprovada

Self-healing workflow ([.github/workflows/self-healing.yaml](#)) demonstrado: - **Execução real:** rollout restart completo em **43 segundos** - Run ID: [#26662680566](#) - **Discord notifica início + sucesso** automaticamente

Cadeia de redução de MTTR:



2. FinOps — Otimização Financeira

Documento completo em [docs/FINOPS-REPORT.md](#) . Resumo:

Tags obrigatórias aplicadas via `default_tags` no Terraform

```

default_tags {
  tags = {
    Project      = "SolidaryTech"
    Environment = "Production" # ou "DR"
```

```
CostCenter = "NGO-Core"
ManagedBy = "Terraform"
Repository = "marcelosilva-dev/tech-challenge-5-fiap"
}
```

Cobertura: 100% dos recursos Terraform-gerenciados (RDS, EKS, ECR, SQS, DynamoDB, VPC, EC2 nodes, EBS volumes).

Rightsizing aplicado

requests / limits configurados em `gitops/<svc>/deployment.yaml` :

Service	CPU req	Mem req	CPU lim	Mem lim
ngo	100m	128Mi	500m	512Mi
donation (Hot Path)	200m	128Mi	1000m	512Mi
volunteer	100m	128Mi	500m	512Mi

Forecast mensal

- **Sempre-on:** ~\$235/mês
- **Uso real (8h/dia × 22 dias):** ~\$74/mês (-70% via `destroy-all.sh`)

Recomendação prática nº 1 — DynamoDB Scan → GSI

Volunteer-service usa `Scan + FilterExpression` no `GET /volunteers/{ngo_id}` — **O(n)** com o tamanho da tabela.

Refatorando para `Query` com GSI `ngo_id-index` : - Cenário 10k voluntários: Scan = 10k RCUs vs GSI Query = 100 RCUs - **Economia: 99%+** naquele endpoint, qualquer escala

Detalhe técnico e código em [docs/FINOPS-REPORT.md](#) §3.

3. ITSM e AIOps — Gestão Preditiva

AIOps

New Relic Applied Intelligence habilitado (free tier 100GB/mês). Detecta anomalias comportamentais automáticas sem regras manuais:

- Latência atípica para a hora do dia
- Padrões de erro em endpoints raramente afetados
- Correlação automática entre alertas relacionados

ITSM — Ciclo de vida de incidente

Documento completo em [docs/ITSM-LIFECYCLE.md](#) . 6 fases:

- 1. **Deteção** (0-2min): Prometheus + AIOps + AWS Health
- 2. **Triagem** (2-5min): SRE on-call avalia, abre war-room Discord
- 3. **Mitigação** (5-15min): Self-healing automático OU runbook manual
- 4. **Resolução**: critérios objetivos (métricas saudáveis 5min)
- 5. **Comunicação**: stakeholders durante e pós (template em PT-BR)
- 6. **Post-Mortem** (48h depois): blameless, com Five Whys + action items

Routing de alertas (Alertmanager)

```
critical → PagerDuty (incident) + Discord (war-room) + GitHub Actions (self-heal)
warning → Discord apenas
SL0 burn → PagerDuty imediato + Discord + escala SRE
```

Config em [gitops/monitoring/alerting/alertmanager-config.yaml.example](#) .

Evidência empírica (Sprint 5.5)

3 alertas enviados, todos chegando em **ambos** canais: - PagerDuty incidents #4, #5, #6 no service `SolidaryTech-Production` - Discord webhook recebendo mensagens em formato Slack-block - Self-healing workflow #26662680566 demonstrado com 3 pods reiniciados em 43s

4. Multicloud, Segurança e Disaster Recovery

Plano de Continuidade de Negócios (PCN)

Documento executivo em [docs/PCN.md](#) . RTO/RPO formais:

Serviço	RTO	RPO	Como
donation-service (Hot Path)	15 min	5 min	RDS Cross-Region Read Replica + EKS DR escala 1 → 3
volunteer-service	5 min	seg	DynamoDB Global Tables (replicação nativa)
ngo-service	4h	24h	Velero backup diário

Estratégia de DR Prática — IMPLEMENTAMOS AMBAS as opções do PDF

Opção A (Backup): Velero com bucket S3 cross-region - Schedule diário às 03:00 UTC, retenção 30 dias - Bucket `solidarytech-velero-backups-<ACCOUNT>` em us-west-2 - Script: [scripts/install-velero.sh](#) - **Validado:** backup

manual Completed itemsBackedUp: 543

Opção B (Warm Standby): Terraform modular para região DR - [terraform/environments/dr/](#) em us-west-2 - EKS cluster skeleton (1 node) + RDS donation Cross-Region Read Replica - DynamoDB Global Tables (replicação ativa) - Failover via 1 comando: [scripts/dr-failover.sh](#) - DR drill mensal automatizado: [.github/workflows/dr-drill.yaml](#)

Detalhe técnico em [docs/DR-STRATEGY.md](#) .

Segurança

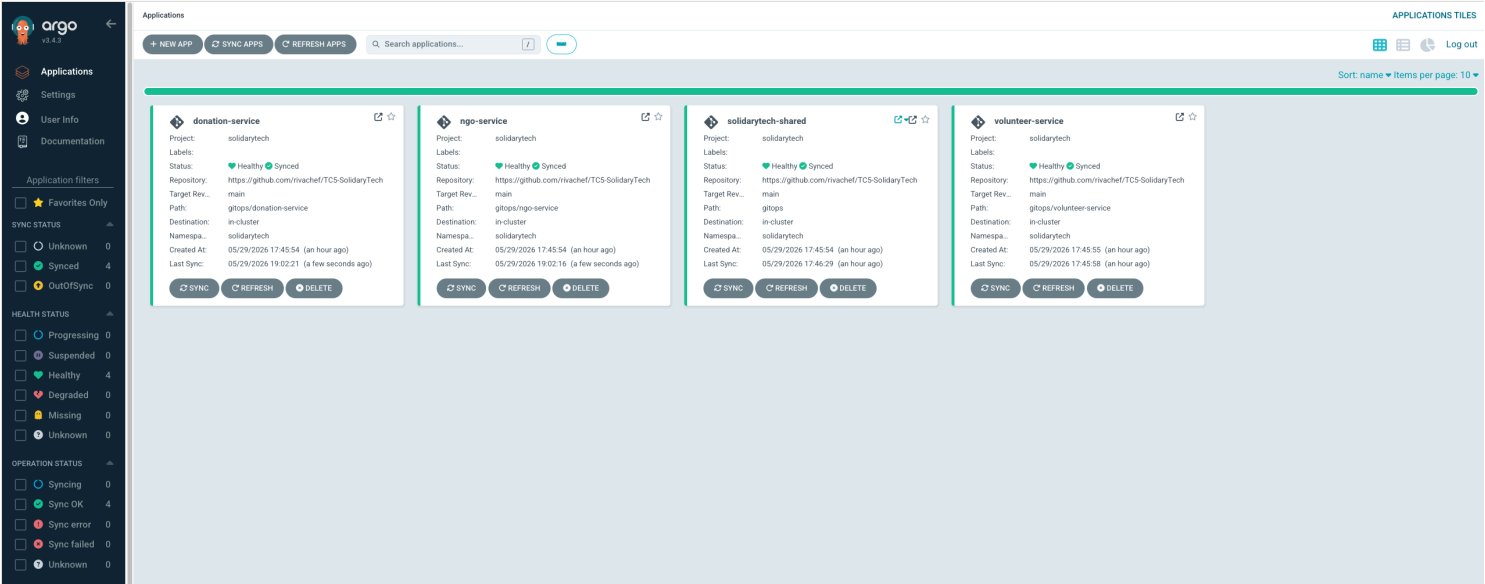
- **DevSecOps em CI:** SAST + SCA + container scan em todos os pushes (item 0)
- **Imagens distroless** no Go (zero CVEs de SO no donation-service)
- **IMDSv2 obrigatório** nos EKS nodes (mitigação SSRF)
- **K8s SecurityContext:** runAsNonRoot: true + allowPrivilegeEscalation: false + drop ALL capabilities
- **Secrets nunca commitados:** .gitignore cobre terraform.tfvars , _generated/ , newrelic-secret.yaml , alertmanager-config.yaml
- **GitGuardian:** alerta tratado durante Sprint 6 — Grafana password movida pra K8s Secret com openssl rand

5. Evidências visuais

Todos os screenshots em [docs/screenshots/](#) . Capturados em 2026-05-29 durante validação E2E do Sprint 7.

5.1 GitOps + ArgoCD

ArgoCD UI — 4 Applications Synced + Healthy :



Prova de que solidaritytech-shared , ngo-service , donation-service e volunteer-service estão totalmente sincronizados com o branch main do Git e em estado Healthy no cluster.

Pods em execução no cluster:

```
=== solidarytech ===
NAME                                READY   STATUS    RESTARTS   AGE
donation-db-init-x5z7m              0/1     Completed 0           3m20s
donation-service-7cdff86d74-b4qfv   1/1     Running   0           93m
donation-service-7cdff86d74-h5r9x   1/1     Running   0           93m
donation-service-7cdff86d74-qlzht   1/1     Running   0           93m
ngo-db-init-nvl84                   0/1     Completed 0           3m26s
ngo-service-5ffdd4cdb9-p5zf8        1/1     Running   0           94m
ngo-service-5ffdd4cdb9-t5tcm        1/1     Running   0           94m
volunteer-service-6485dff57c-tz7gr  1/1     Running   0           125m
volunteer-service-6485dff57c-wd2gj  1/1     Running   0           125m

=== monitoring ===
NAME                                READY   STATUS    RESTARTS   AGE
alertmanager-prometheus-kube-prometheus-alertmanager-0  2/2     Running   0           64m
loki-0                                2/2     Running   0           123m
loki-canary-br64j                      1/1     Running   0           123m
loki-canary-rf7xl                      1/1     Running   0           123m
loki-canary-tjrb9                      1/1     Running   0           123m
otel-collector-opentelemetry-collector-5764544778-9lhpm  1/1     Running   0           122m
prometheus-grafana-769f4c587f-6x9v7    3/3     Running   0           124m
prometheus-kube-prometheus-operator-6f49ffc5c5-fk7f7    1/1     Running   0           124m
prometheus-kube-state-metrics-f56f99797-r5fmj            1/1     Running   0           124m
prometheus-prometheus-kube-prometheus-prometheus-0      2/2     Running   0           124m
prometheus-prometheus-node-exporter-5sfkx                1/1     Running   0           124m
prometheus-prometheus-node-exporter-t7wvt                1/1     Running   0           124m
prometheus-prometheus-node-exporter-wj5cl                1/1     Running   0           124m
promtail-7gzcm                                           1/1     Running   0           122m
promtail-8ccjf                                           1/1     Running   0           122m
promtail-m6mjg                                           1/1     Running   0           122m

=== argocd applications ===
NAME                SYNC STATUS   HEALTH STATUS
donation-service    Synced       Healthy
ngo-service         Synced       Healthy
solidarytech-shared Synced       Healthy
volunteer-service   Synced       Healthy
```

5.2 CI/CD com DevSecOps

3 workflows CI verdes (lint + test + SAST + SCA + container scan + push ECR):

Actions

New workflow

All workflows

CI - donation-service

CI - ngo-service

CI - volunteer-service

DR Drill - Monthly

Self-Healing — Pod Recovery

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

Selected workflows

Showing runs from all workflows named CI - ngo-service, CI - volunteer-service, CI - donation-service

3 workflow run results

Workflow

Event

Status

Branch

Actor

fix(ci): substituir golangci-lint por go vet (compat Go 1.26)

CI - donation-service #7: Commit bf5238b pushed by rivachef

main

Today at 12:39 AM

1m 48s

...

fix(ci): golangci-lint v1.64 (Go 1.24) + boto3 1.35.49 (urllib3 2.x)

CI - volunteer-service #4: Commit 14b970a pushed by rivachef

main

Today at 12:30 AM

1m 21s

...

fix(security): bump golang.org/x/crypto + supress bandit B104

CI - ngo-service #3: Commit 605ab2f pushed by rivachef

main

Today at 12:27 AM

1m 3s

...

5.3 Observabilidade + APM

Grafana — SolidaryTech Overview dashboard (parte 1):



Grafana — Golden Metrics + Logs em tempo real (parte 2):

Dashboards > SolidaryTech > SolidaryTech - Ecosystem Health

[Search](#)

+ Cluster Health

+ Microservices

- Logs (Real-Time)

<<
Last 1 hour
>>
🔍
Refresh
↶ Share
Save
⌵

SolidaryTech Logs (Live)

2026-05-29 19:28:00.152 e/1.34"	app-mgp-service component-backend container-mgp-service job-solidarytech/ngo-service node_name=ip-10-0-10-153.ec2.in.	+3	ngo-service-Sffdd4c9b-p5tcm stdout: 10.0.10.153 -- [29/May/2026:22:28:00 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:59.047 46 "-" "kube-probe/1.34"	app-volunteer-service component-backend container-volunteer-service job-solidarytech/volunteer-serv. node_name=ip-10-0-10-153.ec2.in.	+3	volunteer-service-6485dff57c-wd2gj stdout: 10.0.10.153 -- [29/May/2026:22:27:59 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:59.007 46 "-" "kube-probe/1.34"	app-volunteer-service component-backend container-volunteer-service job-solidarytech/volunteer-serv. node_name=ip-10-0-11-129.ec2.in.	+3	volunteer-service-6485dff57c-tz7gr stdout: 10.0.11.129 -- [29/May/2026:22:27:58 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:56.145 46 "-" "kube-probe/1.34"	app-volunteer-service component-backend container-volunteer-service job-solidarytech/volunteer-serv. node_name=ip-10-0-10-153.ec2.in.	+3	volunteer-service-6485dff57c-wd2gj stdout: 10.0.10.153 -- [29/May/2026:22:27:56 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:55.699 46 "-" "kube-probe/1.34"	app-volunteer-service component-backend container-volunteer-service job-solidarytech/volunteer-serv. node_name=ip-10-0-11-129.ec2.in.	+3	volunteer-service-6485dff57c-tz7gr stdout: 10.0.11.129 -- [29/May/2026:22:27:55 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:58.998 e/1.34"	app-mgp-service component-backend container-mgp-service job-solidarytech/ngo-service node_name=ip-10-0-11-129.ec2.in.	+3	ngo-service-Sffdd4c9b-p5zf8 stdout: 10.0.11.129 -- [29/May/2026:22:27:58 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:58.152 e/1.34"	app-mgp-service component-backend container-mgp-service job-solidarytech/ngo-service node_name=ip-10-0-10-153.ec2.in.	+3	ngo-service-Sffdd4c9b-p5tcm stdout: 10.0.10.153 -- [29/May/2026:22:27:58 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:46.147 46 "-" "kube-probe/1.34"	app-volunteer-service component-backend container-volunteer-service job-solidarytech/volunteer-serv. node_name=ip-10-0-10-153.ec2.in.	+3	volunteer-service-6485dff57c-wd2gj stdout: 10.0.10.153 -- [29/May/2026:22:27:46 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:45.699 46 "-" "kube-probe/1.34"	app-volunteer-service component-backend container-volunteer-service job-solidarytech/volunteer-serv. node_name=ip-10-0-11-129.ec2.in.	+3	volunteer-service-6485dff57c-tz7gr stdout: 10.0.11.129 -- [29/May/2026:22:27:45 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:45.524 e/1.34"	app-mgp-service component-backend container-mgp-service job-solidarytech/ngo-service node_name=ip-10-0-11-129.ec2.in.	+3	ngo-service-Sffdd4c9b-p5zf8 stdout: 10.0.11.129 -- [29/May/2026:22:27:45 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:45.331 e/1.34"	app-mgp-service component-backend container-mgp-service job-solidarytech/ngo-service node_name=ip-10-0-10-153.ec2.in.	+3	ngo-service-Sffdd4c9b-p5tcm stdout: 10.0.10.153 -- [29/May/2026:22:27:45 +0000] "GET /health HTTP/1.1" 200 40 "-" "kube-probe/1.34"
2026-05-29 19:27:41.064 app-donation-db-init container-psql job-solidarytech/donation-db-in.	node_name=ip-10-0-11-129.ec2.in. pod=donation-db-init-x6b65	+2	donation-db-init-x6b65 stdout: Schema aplicado com sucesso.
2026-05-29 19:27:41.061 app-donation-db-init container-psql job-solidarytech/donation-db-in.	node_name=ip-10-0-11-129.ec2.in. pod=donation-db-init-x6b65	+2	donation-db-init-x6b65 stdout: CREATE TABLE
2026-05-29 19:27:41.061 app-donation-db-init container-psql job-solidarytech/donation-db-in.	node_name=ip-10-0-11-129.ec2.in. pod=donation-db-init-x6b65	+2	donation-db-init-x6b65 stderr: psql/sql/init.sql:8: NOTICE: relation "donations" already exists, skipping
2026-05-29 19:27:41.022 app-donation-db-init container-psql job-solidarytech/donation-db-in.	node_name=ip-10-0-11-129.ec2.in. pod=donation-db-init-x6b65	+2	donation-db-init-x6b65 stdout: Aplicando schema donation.db...
2026-05-29 19:27:41.022 1	app-donation-db-init container-psql job-solidarytech/donation-db-in.	+2	donation-db-init-x6b65 stdout: DNS OK (solidarytech-donation-db.c1ubdesikso.us-east-1.rds.amazonaws.com) na tentativa
2026-05-29 19:27:41.009 e/1.34"	app-mgp-service component-backend container-mgp-service job-solidarytech/ngo-service node_name=ip-10-0-11-129.ec2.in.	+3	ngo-service-Sffdd4c9b-p5zf8 stdout: 10.0.11.129 -- [29/May/2026:22:27:40 +0000] "GET /health HTTP/1.1

New Relic APM — lista dos 3 microsserviços:

new relic

Integrations & Agents

All Capabilities

Catalogs

WORKSPACE

Dashboards

APM & Services

Logs

Traces

Synthetic Monitoring

Alerts

Infrastructure

Kubernetes

Browser

Mobile

Errors Inbox

Apps

Find what you need

APM & Services

Save view

Search by entity name

Entity Type

All

Team

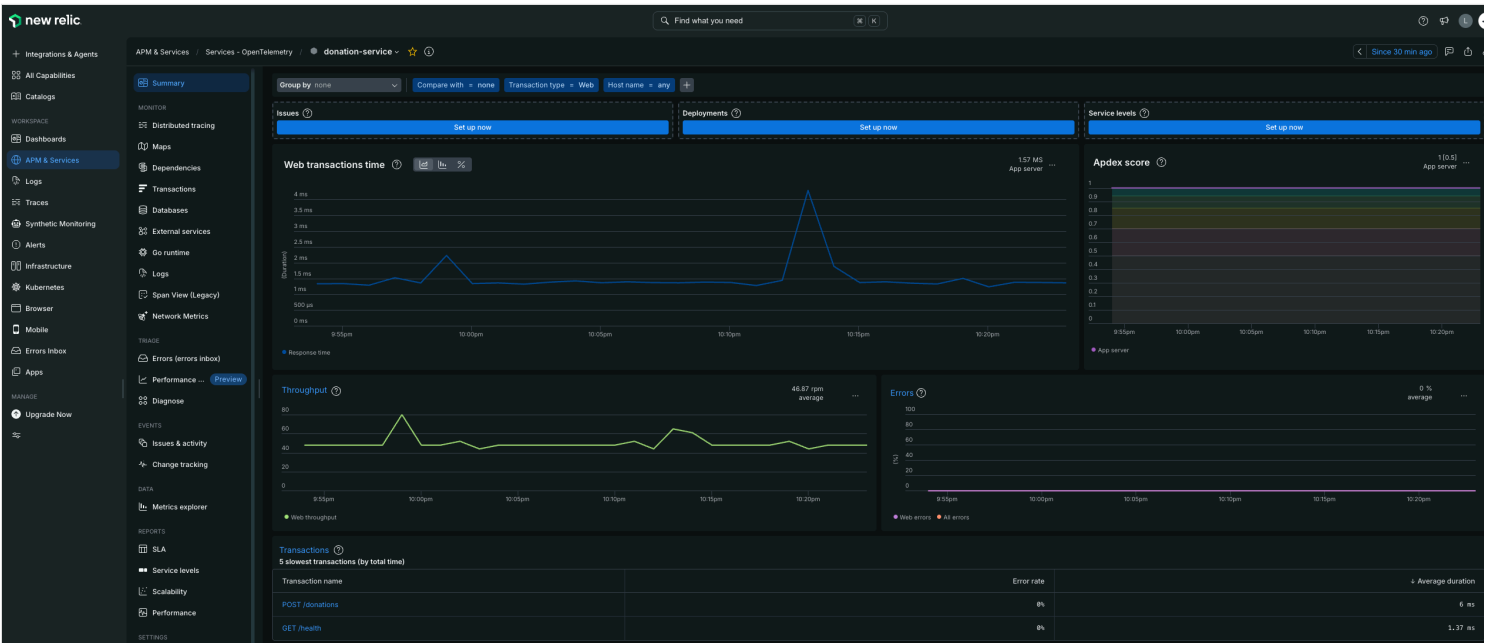
All

+

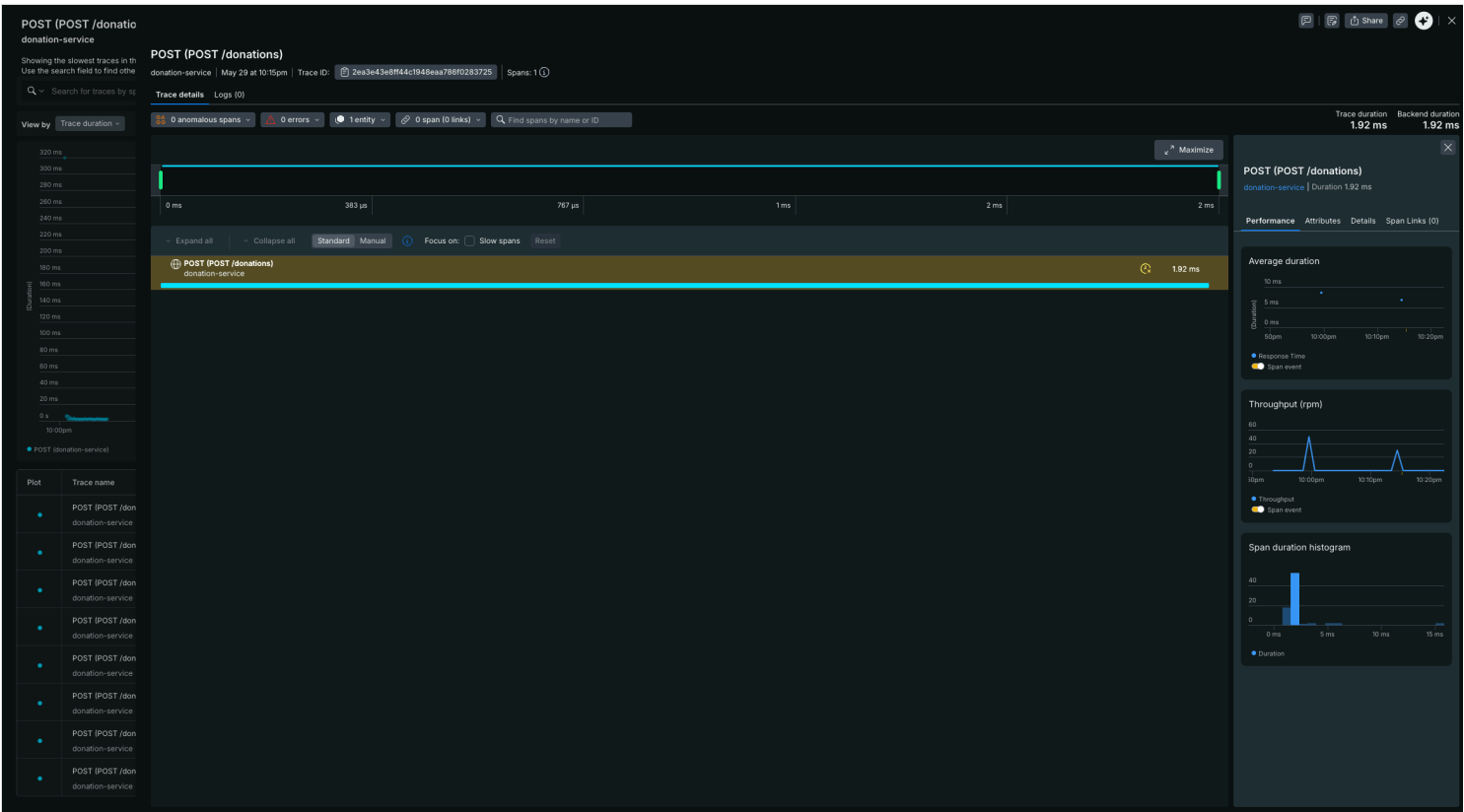
Entities

	Name ↑	Provider	Response time	Throughput	Error rate
☆	● donation-service	opentelemetry	2 ms	46.9 rps	0% ...
☆	● ngo-service	opentelemetry	1 ms	17.9 rps	0% ...
☆	● volunteer-service	opentelemetry	2 ms	17.1 rps	0% ...

New Relic — Summary do donation-service (Hot Path):



New Relic — Distributed Trace waterfall:



Visualiza o caminho completo de uma requisição: HTTP → DB insert → SQS publish, com latência por span.
Base para diagnóstico rápido de degradação.

5.4 ITSM — Gestão de Incidentes

PagerDuty — 3 incidents triggered via Events API V2:

PagerDuty

IncidentsServicesPeopleAutomationAIOpsAnalyticsIntegrationsStatusAI

Search

Your open incidents (Last 30 days)

3 triggered0 acknowledged

All open incidents (Last 30 days)

3 triggered0 acknowledged

AcknowledgeReassignResolveSnooze

Go to incident #...

OpenTriggeredAcknowledgedResolvedAny Status

Assigned to meAllLast 30 days

	Status	Urgency	Type	Title	Created	Service	Assigned To
☐	Triggered	High	Base Incident	[CRITICAL] PostRestart_1780091226 — donation-service SHOW DETAILS (1 triggered alert)	#7 at 6:47 PM	SolidaryTech-Production	Cesar Paulino
☐	Triggered	High	Base Incident	[CRITICAL] E2E_AlertmanagerFlow — donation-service SHOW DETAILS (1 triggered alert)	#6 at 6:45 PM	SolidaryTech-Production	Cesar Paulino
☐	Triggered	High	Base Incident	TESTE POS-FREE-PLAN: deveria aparecer no PagerDuty agora SHOW DETAILS (1 triggered alert)	#5 at 6:44 PM	SolidaryTech-Production	Cesar Paulino

Per Page:

25

<1-3>

Service SolidaryTech-Production com 3 incidents triggered pelo Alertmanager + curl manual de teste. Activity feed mostra "Triggered through the API".

Discord — canal com alertas Alertmanager + notificações self-healing:

TC5 - SolidaryTech

TC5 - SolidaryTech

Eventos

Impulsos de servidor

alerts

Canais de Texto

geral

Canais de Voz

Geral

alerts

Capturing logs in real time

CRITICAL: ExplicitEnd_1780089712
Alert: ExplicitEnd_1780089712 - CRITICAL Summary: Alerta com endsAt explicito 1h futuro Description: Forcar fire prolongado

WARNING: PodNotReady
Alert: PodNotReady - WARNING Summary: Pod nao pronto: solidaritytech/donation-db-init-kkgtn Description: Pod nao atingiu Ready em 10min
Alert: PodNotReady - WARNING Summary: Pod nao pronto: solidaritytech/ngo-db-init-btlbf Description: Pod nao atingiu Ready em 10min

CRITICAL: LiveTest_1780089629
Alert: LiveTest_1780089629 - CRITICAL Summary: Live debug test Description: Capturing logs in real time

Alertmanager APP 18:45

CRITICAL: E2E_AlertmanagerFlow
Alert: E2E_AlertmanagerFlow - CRITICAL Summary: Validacao E2E: Alertmanager -> PagerDuty + Discord Description: Se voce ve este alerta tanto no PagerDuty quanto no Discord, a stack ITSM da SolidaryTech esta operacional.

CRITICAL: PostRestart_1780091226
Alert: PostRestart_1780091226 - CRITICAL Summary: Alerta APOS restart do Alertmanager Description: Se chegar no PagerDuty + Discord, fluxo E2E confirmado.

18:47

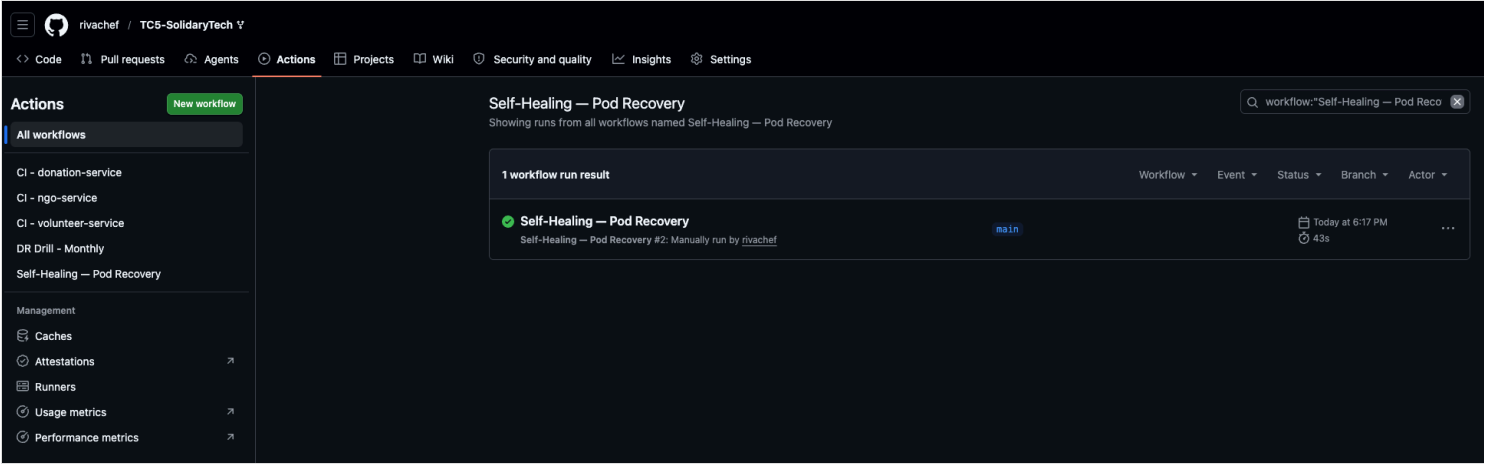
InfoInhibitor
Alert: InfoInhibitor - none Summary: Info-level alert inhibition. Description: This is an alert that is used to inhibit info alerts. By themselves, the info-level alerts are sometimes very noisy, but they are relevant when combined with other alerts. This alert fires whenever there's a severity="info" alert, and stops firing when another alert with a severity of 'warning' or 'critical' starts firing on the same namespace. This alert should be routed to a null receiver and configured to inhibit alerts with severity="info".

Alertmanager APP 18:52

InfoInhibitor
Alert: InfoInhibitor - none Summary: Info-level alert inhibition. Description: This is an alert that is used to inhibit info alerts. By themselves, the info-level alerts are sometimes very noisy, but they are relevant when combined with other alerts. This alert fires whenever there's a severity="info" alert, and stops firing when another alert with a severity of 'warning' or 'critical' starts firing on the same namespace. This alert should be routed to a null receiver and configured to inhibit alerts with severity="info".

5.5 Self-Healing — redução de MTTR

GitHub Actions — workflow self-healing executando rollout restart:



Run #26662680566 — 43 segundos para concluir rollout restart de 3 pods do `donation-service` com notificações Discord automáticas.

5.6 Disaster Recovery

Velero — backup completo cross-region (us-east-1 → us-west-2):

```
=== Velero ===
NAME                                READY   STATUS    RESTARTS   AGE
velero-6df7b6d86-g2xd8             1/1     Running   0           14m

NAME                                PHASE    LAST VALIDATED   AGE    DEFAULT
aws-us-west-2                      Available 27s              14m

NAME                                AGE
manual-sprint6-validation           13m
sprint7-screenshot-evidence         13m
```

DynamoDB Global Tables — replica us-west-2 ACTIVE:

```
luizcesarpaulinodacosta@Luizs-MacBook-Pro primary % echo "=== DynamoDB Global Tables status ===" && \
aws dynamodb describe-table --table-name SolidaryTechVolunteers \
--query 'Table.{Name:TableName,Status:TableStatus,Replicas:Replicas[*].[RegionName,ReplicaStatus]}' \
--output table
=== DynamoDB Global Tables status ===
-----
| DescribeTable |
+-----+
| Name | Status |
+-----+
| SolidaryTechVolunteers | ACTIVE |
+-----+
| Replicas |
+-----+
| us-west-2 | ACTIVE |
+-----+
```

5.7 FinOps — Tags em ação

AWS Tag Editor — 22 recursos com tag Project = SolidaryTech :

Resource search results (22)

Export 22 resources to CSVManage tags of selected resources

Choose up to 500 resources for which you want to edit tags.

22 resources are loaded. Load more resources.

Filter resources

Identifier ID	Tag: Name	Service	Type	Region	Tag: Project	Tags
rtb-04c991b632d50dd97	solidarytech-public-rt	EC2	RouteTable	us-east-1	SolidaryTech	6
subnet-0bcd0c61111414c1e	solidarytech-public-us-east-1b	EC2	Subnet	us-east-1	SolidaryTech	8
tcs-solidarytech-tflock-538593325211	(not tagged)	DynamoDB	Table	us-east-1	SolidaryTech	4
lgw-049ec9760870c106a	solidarytech-lgw	EC2	InternetGateway	us-east-1	SolidaryTech	6
sg-0248f9d2ba17c35a1	solidarytech-eks-nodes-sg	EC2	SecurityGroup	us-east-1	SolidaryTech	6
rtb-0e948f0137bbec21b	solidarytech-private-rt-0	EC2	RouteTable	us-east-1	SolidaryTech	6
sg-0df48dfda2c384df9	solidarytech-rds-postgres-sg	EC2	SecurityGroup	us-east-1	SolidaryTech	6
subnet-09899c0a631570ec8	solidarytech-private-us-east-1b	EC2	Subnet	us-east-1	SolidaryTech	8
SolidaryTechVolunteers	SolidaryTechVolunteers	DynamoDB	Table	us-east-1	SolidaryTech	8
nat-049c0a42181138525	solidarytech-nat-0	EC2	NatGateway	us-east-1	SolidaryTech	6
vpc-05c50b5ebbf3a63f	solidarytech-vpc	EC2	VPC	us-east-1	SolidaryTech	6
subnet-079bdadab3b07bd03	solidarytech-private-us-east-1a	EC2	Subnet	us-east-1	SolidaryTech	8
solidarytech-cluster	solidarytech-cluster	EKS	Cluster	us-east-1	SolidaryTech	7
elpalloc-0027c65a6d529239f	solidarytech-nat-elip-0	EC2	EIP	us-east-1	SolidaryTech	6
subnet-0798eac36ddcf58de	solidarytech-public-us-east-1a	EC2	Subnet	us-east-1	SolidaryTech	8
solidary-donations	solidary-donations	SQS	Queue	us-east-1	SolidaryTech	8
solidarytech-ngo-db	solidarytech-ngo-db	RDS	DBInstance	us-east-1	SolidaryTech	8
rds:solidarytech-ngo-db-2026-05-29-20-...	solidarytech-ngo-db	RDS	DBSnapshot	us-east-1	SolidaryTech	8
solidarytech-donation-db	solidarytech-donation-db	RDS	DBInstance	us-east-1	SolidaryTech	8
rds:solidarytech-donation-db-2026-05-2...	solidarytech-donation-db	RDS	DBSnapshot	us-east-1	SolidaryTech	8
solidarytech-rds-subnet-group	solidarytech-rds-subnet-group	RDS	DBSubnetGroup	us-east-1	SolidaryTech	6
solidary-donations-dlq	solidary-donations-dlq	SQS	Queue	us-east-1	SolidaryTech	8

Cobertura: EC2 (subnets, VPC, IGW, NAT, SGs, route tables, EIP), DynamoDB (volunteers + tflock), EKS (cluster + node group), RDS (2 DBs + 2 snapshots + subnet group), SQS (main + DLQ). Inclui snapshots automáticos do RDS atestando `copy_tags_to_snapshot = true`.

6. Recursos / Links

Recurso	Link
Repositório principal	https://github.com/marcelosilva-dev/tech-challenge-5-fiap
Upstream original	https://github.com/douglshackathon-DCLT
Vídeo de demonstração	(adicionar)
ArgoCD UI	(LoadBalancer URL — dinâmica)
Grafana	(LoadBalancer URL)
New Relic dashboard	https://one.newrelic.com

7. Stack de tecnologias

Categoria	Tecnologias
Cloud	AWS (EKS, RDS PostgreSQL, DynamoDB Global Tables, SQS, ECR, S3, NAT, EBS)
Container	Docker, distroless (Go), python:3.12-slim
Orchestration	Kubernetes 1.34, ArgoCD
IaC	Terraform 1.5+ (modular, multi-environment)
CI/CD	GitHub Actions com Trivy + gosec + bandit + flake8 + golangci-lint
Observability	Prometheus, Grafana, Loki, Promtail, OpenTelemetry Collector
APM	New Relic (OTLP)
ITSM	PagerDuty (Free), Discord webhook, GitHub Actions (self-healing)
AIOps	New Relic Applied Intelligence
DR	Velero (Opção A), Terraform Warm Standby (Opção B), DynamoDB Global Tables
Linguagens	Go 1.26 (Hot Path), Python 3.12 (ngo + volunteer)

8. Métricas do projeto

- **Sprints concluídas:** 7 (1 → 5 + 5.5 + 6)
- **Commits no repositório:** 25+ commits incrementais com mensagens detalhadas
- **Bugs descobertos e corrigidos** (com causa raiz documentada): 20+
- **Linhas de código adicionadas:** ~3.500 (Terraform + YAML + scripts + Go/Python)
- **Documentos técnicos:** 5 (PCN, DR-STRATEGY, ITSM-LIFECYCLE, SRE-SLO, FINOPS-REPORT) + este

Apêndices

- [PCN.md](#) — Plano de Continuidade de Negócios (executivo)
- [DR-STRATEGY.md](#) — Estratégia técnica de DR
- [SRE-SLO.md](#) — SLI/SLO/SLA formal
- [FINOPS-REPORT.md](#) — Forecast + tags + recomendações
- [ITSM-LIFECYCLE.md](#) — Ciclo de vida de incidente
- [VIDEO-ROTEIRO.md](#) — Roteiro do vídeo de demonstração